

Cooling Benefits of Urban Tree Canopy: A Systematic Review

Yihan Yin , Song Li , Xiaoyi Xing , Xinyi Zhou, Yujie Kang, Qi Hu and Yanjing Li

College of Horticulture and Forestry Sciences, Huazhong Agricultural University, Wuhan 430070, China

* Correspondence: lisong@mail.hzau.edu.cn

Abstract: As an important part of urban ecosystems, trees can effectively alleviate the urban heat island effect. Tree canopies cool and humidify through shading and evapotranspiration, regulating the urban thermal environment. So far, many studies have analyzed the heat mitigation effect of urban green spaces; however, there are relatively few studies on the cooling effect of tree canopies. Specifically, relevant research focusing on different spatial scales has not been explored. Therefore, this review systematically summarizes the research on the benefits of tree canopy cooling carried out in recent years, analyzes the research content, and evaluates the indicators and key influencing factors of the benefits of tree canopy cooling from four different spatial scales: urban, block, community, and individual. It was found that canopy factors, other vegetation factors, and environmental factors jointly affect the benefits of canopy cooling. This research focuses on the benefits of canopy cooling at different spatial scales. The smaller the research scale, the more discussion and attention will be paid to vegetation factors. This paper puts forward major directions for future research and development, providing optimization strategies for urban planning or plant design at different scales in the context of climate change.

Keywords: urban heat island; urban green space; tree canopy; cooling effects; thermal comfort



Citation: Yin, Y.; Li, S.; Xing, X.; Zhou, X.; Kang, Y.; Hu, Q.; Li, Y. Cooling Benefits of Urban Tree Canopy: A Systematic Review. *Sustainability* **2024**, *16*, 4955. <https://doi.org/10.3390/su16124955>

Academic Editors: Francesca Ugolini and David Pearlmutter

Received: 16 April 2024

Revised: 31 May 2024

Accepted: 6 June 2024

Published: 10 June 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Human activities and changing land cover types caused by urbanization lead to a series of problems, such as global warming, frequent extreme climate events, and the urban heat island effect (UHI) [1–3]. Urban high temperatures caused by the UHI effect affect the thermal comfort of the human body and even cause a series of health problems, such as heat stroke, breathing difficulties, heat cramps, dehydration, and so on [4]. The UHI effect also raises night temperatures, increasing electricity demand and building energy consumption for cooling [5–7]. These problems have seriously affected human health and limited the sustainable future development of cities. The rising global temperature and the new round of urbanization will lead to the continuous aggravation of thermal problems in urban environments.

A large number of studies have shown that urban green spaces can mitigate the negative effects of the UHI effect at both macro and micro scales [8], and trees are the key elements of urban green space that exert ecological benefits [9]. The cooling benefit of trees is mainly related to canopy shading and evapotranspiration, and the cooling benefit of canopy shading is much stronger than evapotranspiration [10]. Studies have shown that urban forests can significantly reduce sensible heat flux and increase latent heat flux, thereby cooling areas [11]. By intercepting the shortwave radiation of the sun, a tree canopy reduces the storage and convection of surface heat, directly reduces the surface temperature, and absorbs long-wave radiation from the surrounding area to regulate the urban thermal environment [12,13]. Canopy evapotranspiration reduces the air temperature and increases the ambient humidity under and around the trees to regulate the canopy microclimate and improve the thermal comfort level of pedestrians. Previous research on the cooling benefits of tree canopies was mainly conducted at the microclimate scale. In recent years, research on urban public spaces, such as blocks and streets, has been continuously promoted, and

even larger-scale research, such as at the urban scope, has begun to take tree canopies as the research object. Studies have shown that the canyon effect at the block scale can significantly affect the cooling efficiency of street trees [10,14]. Some studies have compared the cooling benefits of tree canopies in open public spaces with artificial shade facilities [15,16]. Taking the city square of Pest, Hungary, as an example, it was found that a small tree was more effective in alleviating heat than a low-hanging solar sail [15]. It can be seen that the effect of tree canopies on urban thermal environment regulation is efficient and irreplaceable. The different climate conditions and solar durations between cities, as well as differences in land cover spatial layout and building density caused by different urbanization degrees, affect the effect of tree canopy cooling.

Many previous studies have included analyses of the cooling benefits of urban green spaces and trees. However, relevant literature reviews tend to focus only on a certain type of spatial scale. Although the existing literature review studies have confirmed that various thermal environment indicators, climate factors, and vegetation parameters of relevant studies have obvious scale characteristics and clustering [17,18], the research focus on cooling benefits at different scales has not been summarized, and the division of spatial scales is often fuzzy macro and micro, lacking more detailed classification. Most studies on canopy cooling benefits focus on the microclimate scale, such as the difference in canopy cooling benefits caused by individual and community vegetation parameters, and there is a lack of reviews on canopy cooling benefits on a larger scale, such as street trees and urban tree cover.

Accordingly, this review summarizes the research progress on the cooling benefits of tree canopies, systematically sorts out research methods and research emphasis at four different spatial scales of urban, block, community, and individual explores the enlightenment of urban tree canopy cooling benefits for green space planning and design at different scales, and finally, proposes prospects for future research directions on the benefits of tree canopy cooling.

2. Materials and Methods

2.1. Search and Selection Process

This systematic review of studies on the cooling benefits of urban tree canopies was conducted using the Preferred Reporting Items for Systematic Reviews (PRISMA) method [19,20] (Figure 1).

This review used the Web of Science as an electronic database to search for publications published before January 2024. Web of Science was chosen because it is a major bibliographical database containing reliable and comprehensive ecological literature. A Boolean operator ‘AND’, ‘OR’ computerized search (“tree canopy” OR “urban trees”) AND (“cooling effect” OR “heat island mitigation” OR “thermal comfort”), was conducted to identify articles.

A total of 2217 articles were initially retrieved, and the inclusion and exclusion processes were screened according to the PRISMA process. Duplicates, non-journal publications, articles where the full text is not freely available, and non-English publications were excluded; thus, 482 articles were excluded. Of the remaining 1735 articles, articles whose research area was completely unrelated to this review were excluded, and the remaining 941 articles were excluded. Then, 752 articles were excluded by skimming the titles and abstracts, mostly because of the low relevance and applicability of the research areas to the content of the review. Finally, the remaining articles were further evaluated and screened by reading the full text; of these, 107 articles were excluded, and 82 articles were finally included. The main reasons for exclusion include the fact that the research subject is not on the canopy itself or that the article did not focus on specific ecological benefits.

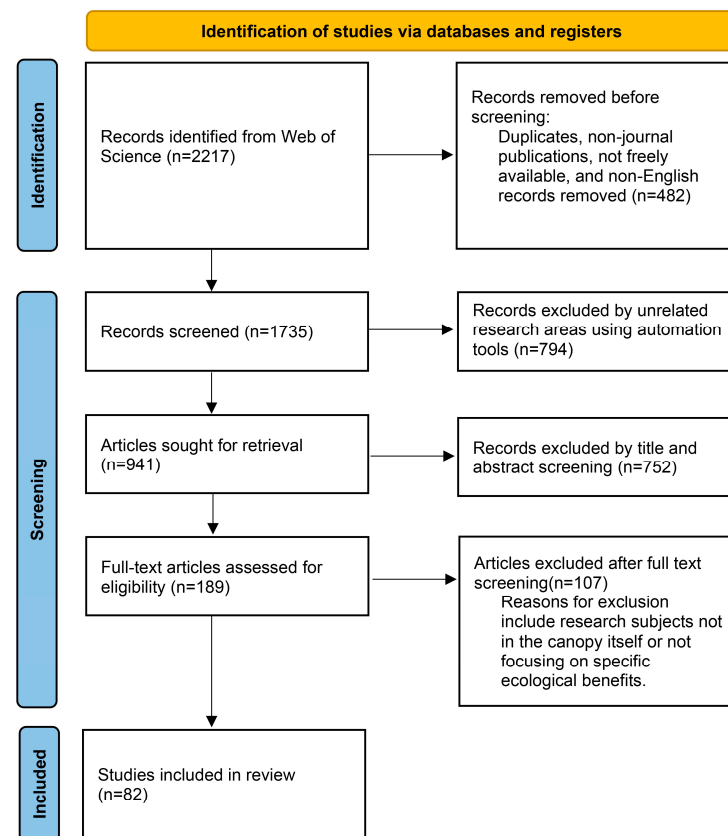


Figure 1. PRISMA flow diagram.

2.2. Spatial Scale Clustering from the Reviewed Studies

By analyzing the results of the review, it was found that the research content relating to urban canopy cooling benefits is obviously clustered (Figure 2) and is mainly distributed across three research fields: (1) In the fields of urban ecology, urban meteorology, and geographic remote sensing, it is necessary to consider canopy cover at urban and even regional scales and study the cooling effect on the urban heat island and urban green space in combination with the local climate and underlying surface composition of the urban area. (2) Architecture: Landscape architecture studies the cooling effect of street trees in the environment of buildings, streets, residential areas, and other blocks and its specific mechanisms, focusing on the thermal comfort of pedestrians. Keywords such as thermal comfort, street tree, and software simulation often appear in this kind of research. (3) Landscape ecology, botany, and other disciplines often study the relationship between community structures, canopy structures, and even the differences in the physiological traits among individual trees and canopy cooling benefits according to the number and structural characteristics of research objects and research perspectives. It can be seen that the differentiated cluster of research focuses on canopy cooling benefits by discussion at different spatial scales.

Based on the differences at different research spatial scales, this paper summarizes the retrieved literature into urban scale, block scale, community scale, and individual scale and summarizes the research characteristics under this scale. Finally, the differences in research methods, research emphases, and experimental methods of studies with different scales are pointed out.

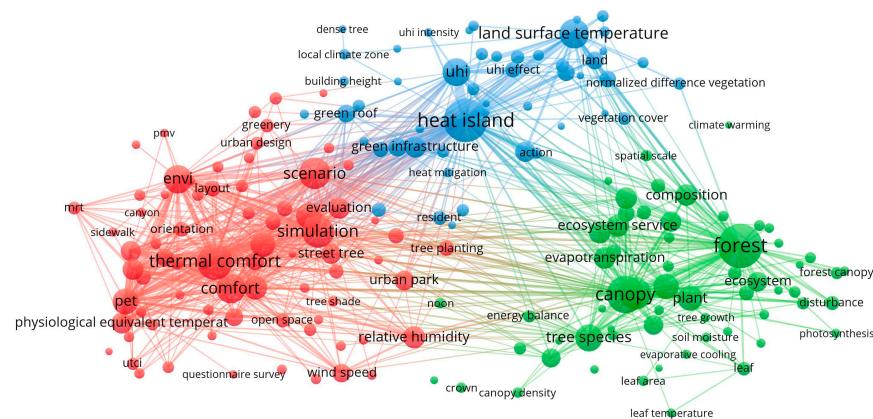


Figure 2. The visualization of keywords clustering network in VOS viewer (version 1.6.20).

3. Results

3.1. Survey Methods for Canopy and Cooling Benefits

3.1.1. Survey Methods for Canopy Parameters

The methods used to assess canopy parameters can be divided into direct and indirect methods. Canopy size (crown diameter, crown width, crown length, etc.), tree height, and crown height can be directly measured using a rangefinder, hypsometers, and other instruments [21,22]. In the case of limited experimental conditions, some parameters can be measured via traditional manual surveying methods, such as in the case of tree height, which can be obtained by comparing shadows, and the leaf area index (LAI), which can be obtained via destructive sampling or the collection of leaf litters and measuring their area [23].

Indirect methods mainly refer to the use of remote sensing images, optical instruments, and laser scanning methods to calculate tree canopy parameters. Two-dimensional canopy parameters, such as green land area and canopy coverage, are easily obtained through the use of remote sensing images. LAI, sky view factor (SVF), and other parameters can be obtained by taking hemispherical images with a fisheye lens and then calculating them using the relevant digital image analyzer or directly measuring them with a canopy analyzer [24–26]. Porosity can be generated by calculating the ratio of the projected area of the transparent pores in the vertical plane of the image to the total projected area of the canopy coverage edge [27]. Three-dimensional data are mainly obtained through 3D laser scanning technology, such as using terrestrial laser scanning (TLS) to measure the vegetation canopy structure and capture the 3D point cloud volume of tree canopies [28]. With the development of drone technology in recent years, obtaining tree canopy parameters through drone-mounted camera scanning is increasingly being used in research in this field [29]. In addition, the transpiration rate of trees is generally evaluated based on the sap flow velocity, which is mainly estimated using real-time measurements of water movement in a single trunk using sap flow sensors [30].

In addition to the survey methods mentioned above, there are often many ways to measure the same parameter. With the development of research, new and more representative indicators representing tree crown characteristics and more convenient parameter acquisition methods are constantly supplementing the field [24]. Researchers selectively use the corresponding parameter types and measurement methods according to the emphasis on different research content and realistic conditions.

3.1.2. Survey Method for Cooling Efficiency Parameters

The evaluation methods for cooling benefits mainly include direct cooling and somatosensory cooling. The cooling parameters discussed in direct cooling mainly include land surface temperature (LST), air temperature (AT), etc. When taking human thermal comfort in an urban thermal environment as the research object, physiological equivalent

temperature (PET), mean radiation temperature (MRT), and universal thermal climate index (UTCI) are often used as quantitative indicators. Among them, PET and UTCI are related to people's perception of heat and can be calculated using the relevant heat balance model [14,31]. PET reflects the ambient air temperature corresponding to the maintenance of body and surface temperature and reflects the relationship between the energy balance of the human body and the surrounding environmental meteorological parameters [32]. UTCI is the equivalent ambient temperature based on various weather and climate conditions, in which the reference personnel in the reference environment produce the same physiological dynamic response as the real environment [33]. In addition, there are also survey methods for evaluating thermal comfort based on subjective thermal response, such as the thermal sensation vote (TSV), the thermal comfort vote (TCV), etc., which are generally obtained via questionnaire surveys combined with a semantic scale. They have been proven to be strongly correlated with PET and other indicators [34,35]. The thermal neutral temperature and thermal acceptability range of different regions, seasons, and populations can be obtained by using this method [36–38].

Large spatial scale studies often use ArcGIS to quantify the spatiotemporal patterns of tree shade cover [39], which is combined with buildings and street shade facilities to construct a “shadow map” of the whole city [40]. Various pieces of numerical simulation software based on computational fluid dynamics (CFD) models are often used to study the effects of different factors on tree canopy cooling at small- and medium-sized scales. CFD-model-based software commonly used in this research field includes FLUENT, PHOENICS, ENVI met, etc. FLUENT and PHOENICS are used to simulate and analyze physical problems, such as airflow and heat exchange, in complex collection areas (such as inside and outside buildings), which are often calculated in combination with the RANS (Reynolds-averaged Navier–Stokes) turbulence model, and are mostly used in the study of block-scale areas and covering building elements [41,42]. ENVI-met is an integral 3D modeling software used to simulate the interaction of surface, vegetation, and atmosphere. It uses a three-dimensional non-static physics model that visually sees the interactions between small-scale microenvironments and simulates the effects of vegetation on microclimates, and it is more versatile than other numerical simulation software. Because it can obtain a complete tree model and its accuracy has been proven many times, it is the mainstream software when studying the benefits of canopy cooling [43–45].

The above software models are all distributed parameter models, which means that at least one variable in the model is related to spatial position and can be simulated from a three-dimensional perspective. In addition, some lumped parameter models have also been applied to microclimate simulations, such as SOLWEIG, RayMan, etc. [31,46]. Each variable in a lumped parameter model is independent of the spatial location, and it is considered that variables are uniform in the whole region. It is often combined with distributed parameter model software, such as ENVI-met, in specific research [14,47,48].

3.2. Cooling Benefits and Influencing Factors of Tree Canopy at Different Spatial Scales

3.2.1. Cooling Benefits and Influencing Factors of Tree Canopy at Urban Scale

Different cities have different economic and social conditions, urbanization degrees and natural environments, resulting in different areas and proportions of urban land use and land cover (LULC), which further results in significant diversity in the area proportion, distribution position, and spatial form of trees in different cities. Studies on the cooling effect of tree canopies at the urban scale often take urban forests or the green spaces where the trees are located as the unit and generally only consider the influence of shading on cooling so as to evaluate the mitigation ability of trees on the heat island effect.

The coverage ratio and relative location of different underlying surfaces will affect the cooling efficiency of urban green spaces. The increase in tree cover percentage (TCP) can effectively reduce LST, and this effect is non-linear [49,50]. Impermeable surfaces typically have higher thermal inertia and lower water availability than trees, which means they absorb and store shortwave radiation during the day and output longwave radiation,

thereby exacerbating the UHI effect [51,52]. In fact, there is a competitive relationship between vegetation cover and artificial cover on surface temperature. Liu et al. found that for most cities when the proportion of artificial surfaces exceeds 70%, the impact on surface temperature is greater than that of trees. This inflection point effect is different in different climate zones and cities with different degrees of development [53]. In addition, different spatial arrangements and forms of land cover can significantly increase or decrease LST even when the composition of land cover remains unchanged [54]. A reasonable layout in terms of the relative position of urban green spaces and impervious surfaces avoids the concentration of impervious areas, reducing the spatial accumulation of urban surface temperature, helping urban green space play a better cooling effect so as to improve the overall urban thermal environment [55,56].

On the urban scale, the overall benefits of green spaces where trees are located are often studied using patch units. The type, size, and aggregation degree of patches can all affect the ability of urban green spaces to alleviate the UHI effect. Through the embedding simulation of the WRF Model (The Weather Research and Forecasting Model), Imran et al. found that when the proportion of vegetation patches in each grid unit increased from 20% to 50%, the LST of mixed forests decreased by 0.6 to 3.4 °C at night, mixed shrubs and grasslands decreased by 0.4 to 3.0 °C, and mixed forests and grasslands decreased by 0.6 to 3.7 °C [57]. Greene and Kedron found that larger and more clustered urban canopy patches can more effectively reduce daytime LST, and more regular patch embeddings, such as hexagons, can reduce uncertainty and error [54,58]. Yang et al. argue that the influence of the shape of green spaces on the cooling effect depends largely on their size; when the blue-green space is less than 1 hectare, the compact shape (round or square) will reduce LST more effectively. When the area is larger than 1 hectare, the complex shape of blue-green space has a better cooling effect [59].

The urban spatial form, especially the relative position of the tree canopy to the buildings, greatly affects the shading of the city and where the shade can be enjoyed [39]. From the perspective of urban planning, Aleksandrowicz et al. pointed out that the quality of urban shading can be improved by controlling parameters, such as the number of street trees, the area covered by tree canopies, and the continuity of tree canopy coverage [40]. In addition, if the cooling effect of buildings and tree canopy shadows on horizontal and vertical surfaces is considered, it will be found that shadows significantly alter the thermal effect of the three-dimensional constructed surface of the city [60]. From this, it can be seen that 3D urban shading models at large spatial scales are gradually being established, and canopy shading will be an important influencing factor for future thermal adaptation urban planning and included in urban planning solutions.

The combination of spatiotemporal patterns at the urban scale for canopy cooling is also a research hotspot. In terms of seasons and weather, studies have shown that urban forest cooling is most pronounced in summer and sunny days, while it is weaker in winter and cloudy conditions [61]. Some scholars have pointed out that the shading effect of tree canopy improves summer thermal comfort but may have adverse effects on cities located in temperate or sub-polar climate zones during winter [62]. In addition, the ability of tree canopies to cool the surrounding environment varies at different times of the day, but the specific effect varies between different cities [54,63]. Taking Tampa and New York City in the United States as examples, tree shade has the strongest cooling effect on LST at 7:30 am [39].

Overall, city-scale studies provide valuable insights into mitigating urban heat island effects through the use of canopy cover. However, due to the background climate, degree of urbanization, and landscape heterogeneity of cities in different geographical regions, research conclusions from a certain region are not widely applicable.

3.2.2. Cooling Benefits and Influencing Factors of Tree Canopy at the Block Scale

Research on the cooling benefits of tree canopies at the block scale focuses more on the regulatory effect of tree canopies on the thermal comfort of pedestrians and analyzes the

influence of block composition characteristics such as tree feature factors, street structure, tree planting layout, building form, and density on canopy cooling benefit.

Street Structure: Effect of Street Tree Cooling Efficiency on Street Canyon

A street canyon refers to a narrow urban street space with continuous and tall buildings on both sides. Street height-to-width ratio, SVF, and other street indices are important indicators when evaluating the degree of street canyons. The street canyon effect refers to airflow movements similar to those of a canyon that form in narrow areas with densely spaced buildings, and there are significant differences in the degree of street canyons with different layouts. Research has shown that the cooling effect of street tree canopies in street canyons is affected [10], mainly because the street canyon effect can cause changes in wind speed, wind power, and wind direction, and the degree of this effect varies greatly among different street canyons [64].

Research shows that people only experience local evaporative cooling effects at low wind speeds, while vegetation generally absorbs more heat from the airflow at higher wind speeds. Therefore, tree transpiration can generate higher cooling benefits in areas with higher wind speeds [65]. This study found that as the depth of street canyons increased, the effect of trees on wind speed and cooling effects was limited. This may be because buildings are taller in deeper canyons, impeding the potentially positive effects on trees in terms of wind speed and airflow patterns [66]. Similar conclusions appear in Morakinyo et al.'s study, which shows that shallow canyons are susceptible to harsher thermal conditions than deeper canyons with similar aspect ratios. Therefore, planting tall trees with a lower canopy density with higher trunks in deeper canyons and vice versa in shallower canyons and open areas can effectively improve the efficiency of tree cooling [14]. From the perspective of relative scale, the magnitude of the cooling effect of trees varies with the tree scale relative to the canyon scale, and trees with larger and denser relative scales have significantly stronger cooling effects [11]. In addition, it was found that in street canyons with different SVFs, trees with different canopy characteristics and trunk heights had different temperature-regulating abilities, especially during the daytime. This is mainly due to the shading or shadow-casting effects of buildings during the day, which outweigh the shading effects of trees [14]. To sum up, planting suitable trees in street canyons of different degrees is more conducive to improving the thermal regulation capacity of trees [67].

From the perspective of block types, different types of blocks have significant differences in terms of canyon degree, building materials, and underlying surface properties. Compared to commercial sites with lower SVFs and surface temperature, planting trees in residential and mixed-use sites with higher SVFs can improve pedestrian comfort by reducing MRT and PET, but commercial sites receive lower cooling benefits from street trees [68].

Spatial Layout: How the Spatial Arrangement of Trees Affects the Thermal Environment of the Block Space?

Many studies have shown that the spatial planting layout of trees has a significant impact on environmental temperature. Effectively designing the relative position and arrangement of trees in specific urban spatial locations can significantly improve the outdoor microclimate and human thermal comfort. Due to the fact that the spatial layout of trees first affects the wind of the environment, tree layouts need to be selected based on different spatial types and climate adjustments. Taking the "Silicon Waha" business Park in Alexandria, Egypt, as an example, the numerical simulation found that randomly placing trees in the central area of the site and equidistant trees around the site could effectively increase the wind speed in the site, thus affecting thermal comfort [69]. Based on a simulation, Zhao et al. found that equal-interval arrangements provide the greatest thermal comfort benefits for blocks in desert areas, followed by clustered arrangements without overlapping tree canopies [45]. Zhang et al. used the height-to-distance ratio of trees (as "Aspect ratio of trees", ART) as an evaluation indicator and found that with the

same number of trees, trees with $ART < 2$ should be the preferred choice to alleviate cold winter and hot summer environments, while trees with $ART \geq 2$ weakened the cooling effect of PET. This is because the overlapping tree canopies resulted in lower-than-expected tree coverage [70].

Another important element affecting canopy cooling efficiency is the relative layout of trees and buildings. Building shading has an inhibitory effect on the efficiency of canopy shading [71], and the degree of inhibition increases with the increase in LAI [14]. Jiao et al. took neighborhoods in Beijing as examples and found through model simulation that in the case of demolishing buildings (i.e., only trees), trees provide 2.28 times more shade than when coexisting with buildings, or even 1.5 times more shade than buildings. The average shading efficiency of trees in Beijing neighborhoods is 43.6% (7.5–72.7%), which means that more than half of the shade of trees is replaced by the shade of buildings [72]. Therefore, planting trees in areas with less building shade is more conducive to unleashing the cooling capacity of trees. It can be seen that optimizing the layout of buildings and trees in the block is one of the effective ways of improving the shading potential of the tree canopy. However, since the spatial spillover of the shadow cooling effect has not been quantitatively tested, the research methods for modeling the statistical relationship between LST, buildings and tree shade are still to be developed.

In addition, research has shown that tree canopies also have a cooling effect on buildings, and their impact on building energy consumption follows the order of layout > canopy height > LAI [73]. Therefore, optimizing the spatial layout of trees also has a positive impact on saving building energy consumption and reducing carbon emissions [6,74]. From the relative position between the tree canopy and the building, the planting direction and relative distance of the trees will affect the energy consumption of a building. The distance between the canopy and a wall is a key parameter that affects the shading and cooling benefits of trees. At a distance of three times the canopy radius from the trunk, up to 24% of solar energy can be reduced [75,76]. Studies have shown that in Sacramento, USA, average annual cooling costs can be saved by USD 14 per tree [77], and the orientation of trees relative to windows and walls affects the energy-saving effect of trees [78]. The high coverage of tree canopies on open structures such as doors and windows during noon time (11:50–15:00) will effectively reduce the thermal impact of solar radiation [79]. Ogueke et al. found through experiments that solar heat mainly enters the building envelope and surrounding environment from the west and north-facing surfaces of the building [75], which is consistent with multiple studies showing that planting trees on the west side of a building provides the greatest energy savings [80–82]. From the perspective of the relative height of trees and buildings, Balter et al. conducted thermal analysis and resident interviews on apartments with building envelopes above and below the canopy to calculate PMV. The results showed that apartments under the canopy fully utilized the microclimate effect generated by the canopy, thereby saving air conditioning energy consumption in winter and summer [83]. This also indirectly reflects the cooling effect of tree canopies, which is more effective for buildings below the height of the canopy [84]. It follows that tree canopies play a significant role in building cooling and energy saving. More and more studies are devoted to directly calculating the energy-saving and emission-reducing abilities of tree canopies through software simulations. Research on how to optimize the relative spatial layout and quantify the effects of natural climate conditions in different regions, building height, facade form, and materials on the potential of tree canopy cooling and energy saving is the focus of this field.

Through this review, it was found that studies at the block scale may be more inclined to study the impact of a single factor on the street thermal environment, such as improving the tree canopy factor on street thermal comfort because this can more directly assess and quantify the impact of the factor. The coupling effect of landscape factors and tree characteristics on the thermal environment in the street area is still not well understood. At present, there may be a lack of adequate research methods to accurately model and analyze the combined effects of multiple factors.

3.2.3. Cooling Benefits and Influencing Factors of Tree Canopy at Community Scale

Research on the cooling benefits of tree canopies at the community scale generally takes the characteristics of community structures such as tree species composition and planting structure as the starting point, considering the impact of tree canopy on the overall microclimate effect of the community.

The microclimate effects generated by the same community configuration modes are different. It was found that urban planting communities with more complex configuration patterns and closer proximity to natural forest forms provide stronger microclimate regulation benefits [85,86]. Research in the urban area of Hong Kong shows that compared to communities with a single high canopy and understory vegetation, communities with multiple layers of trees, shrubs, and herbaceous plants can lower temperatures by 1 °C on sunny days in summer and 0.5 °C on cloudy days [61]. Richards et al. compared the cooling effects of five types of vegetation communities in Singapore cities, including artificially managed tree communities, artificially managed tree and shrub communities, and secondary forests. They found that natural secondary forests had the best cooling effect, while the tree and shrub communities planted under the canopy had better cooling effects than pure tree communities [87]. In terms of community composition, the cooling effect of an arbor–shrub–grass community was better than that of an arbor–grass community or arboreal community [88].

The difference in community canopy structure is one of the main reasons for the different cooling benefits of tree communities. Canopy structure can be divided into three characteristics: canopy density, horizontal canopy structure, and vertical canopy structure [89]. Among them, canopy density can be measured using the leaf area index, porosity, etc. Research has shown that tree canopies with high leaf area densities can effectively absorb solar radiation from the top of the community canopy [65]. Wei et al. found that canopy density is significantly positively correlated with both porosity and cooling effect. When canopy density ranges from 0.81 to 0.85, and porosity ranges from 0.31 to 0.35, the cooling effect of a plant community reaches its peak [27]. From the perspective of a horizontal community structure, the cooling effect of urban tree communities improves with increasing canopy coverage, and the relationship between it and the cooling and humidifying effect brought about by transpiration is non-linear, which means that the maximum transpiration rate may occur under certain canopy coverage [90,91]. A study has found that the average canopy width in the community is positively correlated with the cooling range in summer and autumn, while the density of green trees is negatively correlated with the cooling range in winter [92]. From a vertical structure perspective, canopy height is one of the conditions that determines the amount of shade produced by a tree. Trees with high and wide canopies can create a greater shaded area during the day, but this is not conducive to cooling at night [93]. Compared with monolayer and bilayer plant communities, multilayer plant communities are more effective in cooling and humidifying, mainly because their total leaf area is larger and they can reflect more direct solar radiation [94]. Wang et al. used foliage height diversity (FHD) to quantify the uniformity and diversity of canopy leaf area and found that an uneven distribution of the upper canopy ($FHD < 3$) and a relatively closed canopy structure can provide a strong summer cooling effect [89].

From the perspective of community planting patterns, studies have shown that in terms of the microclimate-regulating abilities of different planting modes, they are ordered as follows: group planting > linear planting > individual planting. This may be due to the overlapping of tree canopies in group planting, which blocks direct sunlight, preventing leaf stomata from closing due to excessive temperature, and the cooling benefits brought about by tree transpiration still play a significant role [85]. For single-row forms of tree planting, increasing the number of tree rows leads to an increase in air temperature and UTCI downstream of the canopy when transpiration has less of a cooling effect (humidity and low temperature) [65]. Cai et al. found that the cooling effect of double-sided planting

is 1.7 times that of single-sided planting, meaning that double-row trees provide better shading during the day than single-row trees [95,96].

In addition, research has confirmed a positive correlation between species diversity within communities and the magnitude of cooling. Chinchilla et al. found a positive correlation between tree diversity in urban forests and surface temperature, with this correlation being more significant during warm and dry periods [97]. Wang et al. found that there is a strong correlation between the decrease in temperature and the Shannon–Wiener index and the species richness of the community, and its impact intensity varies seasonally [92]. On the basis of this study, Rendon and Paola confirmed that the positive effect of tree species diversity on cooling benefits still holds on a larger scale and found that diverse urban forests can buffer the drastic changes in daily temperature. They proposed five possible hypotheses, among which they mentioned that it may be because tree species diversity reflects the pattern of urban forest structure diversification, and multi-layer structures have stronger cooling abilities, or because diversified urban forests have greater resistance to urban stress, etc. [98]. At present, there are still few studies on the correlation between tree species diversity and cooling efficiency in communities; however, this provides a new way of improving the cooling effect of urban green spaces without increasing the area of green space, and it is worth conducting more relevant in-depth research.

Although there are many types of canopy parameters that define community characteristics, the key influencing parameters are still inconclusive. Communities with complex configuration patterns and high tree diversity have been shown to have higher cooling potential, and the specific mechanisms remain to be explored.

3.2.4. Cooling Benefits and Influencing Factors of Tree Canopy at Individual Scale

The cooling benefits of tree canopies at the individual scale focus on a comparative analysis of the correlation between the differences in the biological traits and cooling abilities among tree species. Many studies have shown that interspecific differences are the most important factors affecting tree cooling and regulating thermal comfort [96]. The cooling benefits of different tree species are affected by their own canopy structure characteristics, leaf traits, transpiration rates, and other biological characteristics, as well as the climate and environment.

Canopy structure differences are reflected in canopy size, canopy height, canopy morphology, canopy volume (CV), leaf area, and other aspects. Generally speaking, the larger and taller the canopy of a tree, the stronger its cooling ability, as it can block more solar radiation [28]. Recent research has found that under the same coverage conditions, small-canopy trees have better cooling effects than large-canopy trees [43]. Helletsgruber et al. found that tree species with lower tree height and canopy base height, as well as denser leaves, are beneficial in terms of canopy shading when regulating pedestrian-scale thermal comfort [99]. However, some studies have also pointed out that higher tree trunks cause tree canopies with higher leaf temperatures to be further away from the pedestrian level, contributing to canopy transpiration cooling [65]. Some studies have integrated the ratio of crown radius to trunk height (R/TH) as a parameter index and found that its thermal regulation effect is significant. With an increase of 1 R/TH , PET will decrease by nearly 2.5 °C in the morning and afternoon [100].

From the perspective of canopy morphology, there are significant differences in regulating thermal comfort among tree species with different canopy shapes. Research has shown that trees with wider and higher canopies, such as umbrellas and oblongs, are most effective in improving outdoor thermal comfort. Taking the new residential area in Singapore at 2 pm as an example, every 10,000 m² needs 2.20, 12.20, 29.33, and 36.76 umbrella-shaped, oblong, round, and inverted cone trees to reduce PET by 1 °C [101]. It was found that the correlation between the canopy cooling ability and canopy structure was different in the case of different canopy shapes. For example, tree height, leaf area index, and other factors were significantly correlated with AT under an umbrella canopy but not under a cylindri-

cal canopy [95]. The canopy height of spire-shaped trees had a significant effect on PET compared with other canopy shapes, mainly because their CV and LAI were significantly higher than those of other canopy shapes [102]. Furthermore, the cooling effect of different tree canopies is affected by other tree factors. When tree cover is greater than 50% and canopy diameter is 3 m, the cooling ability of cylindrical canopies is significantly higher than that of conical and ellipsoidal canopies [43].

Studies have found that leaf area is important for the three main cooling mechanisms (transpiration, solar radiation reflection, and shading), and it is also a central feature in regulating thermal comfort [103,104]. Higher leaf area is one of the central characteristics of mature trees that provide greater transpiration cooling [105]. Zheng et al. found a direct relationship between LAI, solar radiation attenuation, and PET benefits by comparing the differences in microclimate characteristics of four tree species [106]. Tree species with higher LAI and larger canopies have a stronger ability to convert sensible heat into latent heat [107].

Concerning the biological and physiological characteristics of trees, the transpiration rates of different tree species are different, the physical, biological, and anatomical structures of leaves in the tree canopies are different, and the anatomical structure of the xylem of the tree species itself will affect the evapotranspiration and cooling abilities of a tree canopy [104,108]. The transpiration rates of different tree species can affect the air temperature and humidity under the canopy, which, in turn, affects human thermal comfort under and around the tree canopy. For example, *Tilia cordata* Mill. provides more than four times the transpiration of *Robinia pseudoacacia* L. and can provide 7 °C lower Δ PET than *R. pseudoacacia* under the same conditions [22]. Taking local trees in Dresden, Germany, as an example, *Corylus colurna* and *Tilia cordata* 'Greenspire', with high leaf area density (LAD) and high transpiration rates, have the highest cooling potential. This study also found that reducing the average air temperature by 0.5 K would require a fourfold increase in the transpiration rate [109].

Concerning the biological characteristics of the leaves themselves, the shape, thickness, and texture of the leaves can all affect the cooling abilities of a tree canopy. In terms of transpiration rate, the most important leaf traits are leaf thickness and shape. For leaf shape, research has found that tree transpiration decreases with increased specific leaf area (SLA). In terms of blade thickness, compared to thick blades, thin blades with a thickness of <0.15 mm exhibit higher AT and ST benefits [104]. However, there is a trade-off between the effects of various leaf attributes on cooling efficiency. For example, deciduous trees often have greater cooling capacities than evergreen trees, mainly because the leaves of evergreen trees are thicker and waxy, with lower stomatal opening and increased stomatal resistance. These physiological characteristics are beneficial for reducing transpiration water consumption and improving transpiration efficiency [110]. Overall, the cooling mechanisms of shading and transpiration should be considered in combination to achieve optimal thermal comfort for humans.

In addition, the process by which trees exert their ability to regulate microclimates is influenced by climate conditions and surrounding environmental factors (underlying surfaces, soil, etc.). The climate conditions of cities in different climate zones result in differences in the ability of trees to provide shade and transpiration cooling [104]. Additionally, the cooling ability of tree canopies varies in different climatic periods in the same region, and the ability to adjust the thermal comfort of the shade in a dry period is significantly higher than in a wet period. This relationship is stronger on grassland surfaces than on hard surfaces, and as atmospheric dryness increases, the difference in latent heat between shaded and sunny grassland surfaces becomes increasingly significant [111]. A similar conclusion appears in Herb's study on surface temperature under tree canopies, where under humid weather conditions, the differences in surface temperature under a canopy are much smaller among different surface types [112].

Underlying surfaces with different material types are one of the factors influencing the ability of tree canopies to provide thermal relief, and the color, permeability, and albedo

of the underlying surface can all affect the thermal comfort under a canopy. Research has found that the transpiration of trees growing on the surface of lawns is 10 times higher than that of trees growing on asphalt roads, which greatly affects their cooling and humidifying abilities [104]. Research has found that on asphalt surfaces, the shading effect of *Tilia cordata* Mill. with a higher canopy density is better than that of *Robinia pseudoacacia* L. However, on grassland surfaces, there is no significant difference in the surface cooling effect between the two tree species. This is because the asphalt surface absorbs a large amount of solar radiation and does not have the same evaporative cooling process as grass [113]. Multiple studies have shown that the combination of cool underlying materials and tree canopies can achieve the maximum thermal comfort regulation effect [114–116]. Taking Putrajaya, Malaysia, as an example, a higher canopy density (LAI 9.7) combined with a “cold” underlying surface material (with an albedo of 0.8) has the greatest cooling effect [114]. Canopy cover can improve ground albedo and reduce thermal comfort [117]. This provides a new direction for combining various urban landscape elements to maximize the cooling benefits of tree canopies (Table 1).

Table 1. Factors affecting canopy cooling efficiency at the individual scale.

Types	Influencing Factors	Dependent Variable	Conclusions	References
Canopy structure factors	Canopy width	AT	The tree canopy exerts its cooling power mainly by blocking solar radiation	Kong et al., 2016 [28]
		AT, relative humidity (RH)	Under the same coverage, the cooling effect of small-canopy trees is better than that of large-canopy trees	Wang et al., 2023 [43]
	Canopy height	AT, wet bulb globe temperature (WBGT), etc.	Tree species with a lower canopy base are beneficial to exerting a canopy shading effect	Helletsgruber et al., 2020 [99]
		UTCI	Tree canopies farther from pedestrian height help transpiration and cooling	Manickathan et al., 2018 [65]
	LAI	PET	There is a direct relationship between LAI and solar radiation attenuation and PET benefits	Zheng et al., 2018 [106]
		Transpiration, solar radiation reflection and shading	LAI is the core feature of regulating human thermal comfort	Smithers et al., 2018 [103]; Rahman et al., 2020 [104]
		The ability to convert latent heat	The higher the LAI and the larger the tree canopy, the stronger the ability to convert sensible heat into latent heat	Wang et al., 2022 [107]
	Canopy morphology	PET	Trees with wider and taller canopies are the most effective in improving outdoor thermal comfort	Liu et al., 2022 [101]
		Correlation between canopy cooling ability and canopy structure	Tree height, leaf area index and other factors were more significantly correlated with AT under an umbrella canopy	Cai et al., 2022 [95]
Biophysiological traits *	Transpiration rate	PET	<i>Tilia cordata</i> Mill. has more than four times the transpiration of <i>Robinia pseudoacacia</i> L. and can provide a lower Δ PET of 7 °C	Rahman et al., 2020 [22]
		AT, PET	Trees with high transpiration rates are more effective in reducing AT and PET	Gillner et al., 2015 [109]
	Leaf shape	-	Tree transpiration decreased with the increase in SLA	Rahman et al., 2020 [104]
	Leaf thickness	AT, ST	Thin blades with thickness < 0.15 mm showed higher AT and ST benefits	
	Leaf texture	-	Thicker waxy leaves have lower water consumption and higher transpiration efficiency	Chen et al., 2019 [110]

Table 1. Cont.

Types	Influencing Factors	Dependent Variable	Conclusions	References
Climate and environmental factors	Climatic zone type	ST, evapotranspiration cooling	The climatic characteristics of cities in different climate zones make trees have different abilities of shading and transpiration	Rahman et al., 2020 [104]
	Atmospheric dryness	Latent heat flux	The thermal comfort adjustment ability of shade in dry period was significantly higher than that in wet period	Rahman et al., 2021 [111]
		Correlation between subcanopy surface temperature and underlying surface type	Under wet weather conditions, subcanopy surface temperatures differ much less between different surface types	Herb et al., 2008 [112]
	Underlying surface type	Soil water content, transpiration rate	Trees growing on the surface of a lawn have 10 times more transpiration than trees growing on asphalt, which affects their ability to cool and humidify	Rahman et al., 2020 [104]
		AT, RH, PET, etc.	The combination of the cool underlying surface material with the canopy allows for maximum thermal comfort adjustment	Shahidan et al., 2012 [114]; Rahman et al., 2019 [113]; Wang et al., 2021 [116]

* The fundamental influencing factor of the biophysiological traits mentioned here is transpiration rate.

Individual-scale studies provide detailed studies of the biological properties that influence canopy cooling, but they often isolate these properties from the broader ecological and climatic context. According to the environmental and climatic characteristics of different geographical regions, it is necessary to carry out ecological service function assessment combined with the thermal adaptation abilities of tree species.

4. Discussion

This review is based on existing research results on the cooling benefits of urban tree canopies and considers the perspectives on regulating urban thermal environments. It summarizes the current research focus and proposes the future directions of this research field according to the existing research gaps.

4.1. Research Focus and Gaps at Different Spatial Scales

4.1.1. Urban Scale: From Macro Quantitative to Precise Modeling

Early studies showed the quantitative relationship between urban vegetation coverage, proportion of land cover types, spatial arrangement, and cooling amplitude on a macro scale [13,118,119]. With the continuous deepening of research in this field and the development of related modeling software, urban shading models that combine buildings, trees, and urban infrastructure are being developed and established at the urban scale. However, the same tree planning strategy will not produce the same benefits in different cities and regions within different climatic zones. Taking cities with Temperate Monsoon Climates (TMCs) and Mediterranean Climates (MCs) as examples, a study found that the larger the tree-covered urban green vegetation (UGV) in the MC zone can affect local microclimate, improving the cooling effect. This may be due to the drier climate of the MC zone, resulting in a more pronounced canopy transpiration cooling effect. In addition, the correlation between wind speed and canopy cooling was found to be relatively weak in cities in the TMC zone. This can be explained by the presence of more high-rise buildings and highly compact urban structures in these TMC cities, and the microclimate environment may be more complex, which, in turn, affects the microclimate effect [120]. It can be seen that urban tree planning needs to consider the characteristics of different climate zones and urban structure characteristics to maximize the investment return ratio of planting trees.

4.1.2. Street Scale: Multivariate Study with Numerical Simulation

Research on the cooling efficiency of tree canopies at the block scale needs to consider various factors, such as aerodynamics and building materials, thus exhibiting the char-

acteristics of interdisciplinary integration. At present, there is abundant research on the impact of a single factor on block thermal environments, but there is still limited research on the combined effects of multiple factors. This is mainly due to the heterogeneity and complexity of the urban street environment, which is affected by many factors, including building layout, materials, vegetation, wind speed, solar radiation, and so on. These factors interact to form a complex system, and studying this complexity requires highly sophisticated numerical simulation models and methods. In recent studies, street trees have been included in various block-scale urban canopy models (UCMs). For example, Krayenhoff et al. proposed the Building Effect Parameterization with Trees (BEP-Tree) model based on UCMs. The model can consider the combined effects of street trees on the block-scale climate in terms of radiation exchange, energy balance, wind resistance, turbulence, and pedestrian-level climate, which is suitable for studying the thermal mitigation potential of street trees at the block scale [121,122].

4.1.3. Community Scale: Screening Key Factors Affecting Community Cooling

When the cooling benefits of tree canopies are measured at the community scale, it is biased towards studying the effects of different factors on the microclimate changes around tree canopy and comprehensively studying the effects of tree canopy structural parameters and other tree factors such as diameter at breast height, tree height, and transpiration rate on the cooling benefits of the community. At present, research on the key factors causing canopy cooling benefits in community configurations still has unclear results, and the correlation between various factors has not been determined yet. This may be because the evaluation of community canopy structure indicators is still being supplemented, and their correlation with canopy cooling is being verified. For example, plant area index (PAI) and LAD, as new plant parameter indicators created on the basis of LAI, are considered to be more suitable for analyzing the thermal comfort improvement potential of trees [10,123]. In addition, in recent years, studies have shown that communities with high species diversity have higher cooling benefits, but the mechanism of how tree diversity supports the cooling effect is unclear. Research on such complex community indicators and their specific mechanisms is of practical value [92].

4.1.4. Individual Scale: Interspecific Differences in Cooling Efficiency of Tree Species to Thermal Adaptation

The study of cooling benefits at the individual scale focuses on the differences in biological traits between species, combined with the differences in the canopy parameters and transpiration rates of tree species. At present, there is a lack of quantitative studies on the contribution of canopy shading and evapotranspiration processes to cooling efficiency, especially evapotranspiration. This is mainly because the mechanism of evapotranspiration cooling is greatly affected by the climate and surrounding environmental factors. Extreme air humidity temperatures, wind speed, and soil moisture content will affect evapotranspiration performance to a great extent [124,125]. Currently, research on the cooling efficiency of single tree species mostly focuses on the comparison of tree species' thermal mitigation abilities, lacking comprehensive analysis combined with tree thermal adaptability. Previous studies have paid more attention to urban trees as a strategy to cope with climate change. In recent years, some researchers have begun to pay attention to the impact of climate change on the health status, overall development, and sustainable succession of urban vegetation. Climate warming aggravates the aging of trees, and the extremely dry and high-temperature climates that exceed the capacity of vegetation lead to stress in terms of growth, thus affecting ecological regulatory functions, such as cooling and carbon sequestration [126,127]. Therefore, focusing only on the cooling capacity of trees while ignoring their capacity for thermal adaptation is not in line with the ultimate goal of sustainable urban environmental development.

4.2. Recommendations for Future Work

Based on the current research focus and gap, future research can consider the following suggestions:

- (1) Construct optimal urban tree planning models based on the adaptability of trees to different locations. Future research needs to summarize and classify cities within each climate zone, establish a vegetation database for cities with similar climate types and plant growth environments, and study the cooling benefits based on the suitability of different trees. For urban built-up environments, it is necessary to construct a vegetation evolution model with a time scale and propose optimization approaches for future time evolution processes. For example, the establishment of an Individual Tree Inventory (ITI) model can realize the accurate assessment and spatial distribution of individual trees in a geographical area. A long-term effective inventory can provide baseline information on tree management and dynamic changes over the years and promote sustainable urban tree planning by helping to make plans and find problems [128,129], which is also applicable to thermal adaptation surveys for trees. Based on the economic and social characteristics and development goals of each city, optimal layouts for tree planning in different land use areas should be developed, and urban greening construction should be carried out according to local conditions.
- (2) Study the thermal comfort benefits of street trees based on the combination of block spatial structures and architectural elements. Future research needs to expand on multiple environmental factors while also considering the combined effects of canopy factors and underlying surface characteristics on thermal comfort under tree canopies. To develop numerical simulation models that include more accurate descriptions of tree characteristics, consider street structure, architectural layout, and plant planting as a whole, and study tree planting patterns that have more thermal mitigation effects based on different street characteristics and block types.
- (3) Establish a correlation study on the cooling benefits of street trees at the “city-street” scale. There is a clear continuity between the block and urban scales, but there are differences in the factors that affect the cooling efficiency of tree canopies at different scales. Currently, thermal environment simulation results at the block scale are difficult to apply to entire cities, and the influence range of the cooling benefits of street tree clusters in a city and the correlation to overall urban heat island effect mitigation are unknown. Therefore, it is necessary to study the correlation between the cooling benefits of street trees at different spatial scales and incorporate them into urban green space patterns. On this basis, a “city-street” street tree network with both local block characteristics and overall urban environment characteristics should be established.
- (4) Screen the key canopy structure parameters that affect community cooling and analyze the correlation between community configuration, species diversity, and cooling benefits. Future research should focus on plant communities, analyze and study the correlation between different canopy factors, further quantify the impact of each factor on canopy cooling, and screen the key parameters that affect tree cooling through comparison. Study the specific mechanism of changes in cooling benefits caused by community diversity in order to optimize the configuration mode of plant communities in urban green spaces with limited green space area and the development directions of community diversity.
- (5) Comprehensively study the cooling mechanism of tree canopies based on urban landscape elements. Future research directions will comprehensively consider environmental factors, such as the tree canopy, underlying surface, and soil, and analyze the effectiveness of tree canopy cooling in combination with urban natural environment and landscape factors. Establish life cycle assessment and management of individual tree species, including thermal adaptation through long-term, region-scale tree inventory data recording. Against the background of global climate change and urbanization, reducing climate change stress on trees and improving the thermal

environment should be cultivated according to local conditions based on different urban natural environmental conditions and green space types in combination with the research results of canopy cooling efficiency and tree climate adaptability at the community scale.

5. Conclusions

This paper reviews the previous studies on canopy cooling benefits and reveals the changes in canopy cooling benefits at different spatial scales (urban, block, community, and individual) and a series of influencing factors from multiple perspectives. The following conclusions can be drawn from this review:

- Canopy factors, other vegetation factors, and environmental factors affect the potential cooling effect of tree canopies to different degrees. The influencing factors of canopy cooling efficiency are different at different spatial scales. The smaller the research scale, the more attention and discussion on vegetation factors.
- At the urban scale, research has focused on considering the overall coverage and distribution of tree canopy cover and how it mitigates the urban heat island effect through shading effects.
- Block-scale studies highlight the impact of street structures and spatial layouts on the cooling effectiveness of street trees, in particular, how the street canyon effect changes wind speed and direction, which, in turn, affects the transpiration and shading effects of trees.
- Community-scale studies reveal the effects of community configuration, community structure, and planting patterns on the overall microclimate effect, indicating that complex configuration patterns and near-natural communities provide stronger microclimate regulatory benefits.
- Individual-scale studies focus on analyzing the correlation between the biological characteristics of different tree species (such as canopy structure, leaf characteristics, and transpiration rate) and cooling capacity and how these characteristics affect thermal comfort around a tree.

Although this review is comprehensive, it has some limitations. The studies included may not be representative of all geographic areas, and the diversity of methods may affect the comparability of the study results. Therefore, we propose the following points to provide some reference for similar literature analysis in the future: (1) Research options should be focused on more diverse geographical areas to better understand the global context of urban canopy cooling benefits. (2) Standardization of data collection and analysis methods should be encouraged to facilitate more meaningful comparisons and aggregation of findings across studies. (3) Longitudinal studies that track changes over time and studies that assess the adaptation of trees to climate change are essential for the sustainable development of urban forestry. By addressing limitations on the basis of these findings, future research could further refine our understanding of the role of urban trees in addressing the effects of climate change.

Author Contributions: Conceptualization, Y.Y., X.X., and S.L.; writing—original draft, Y.Y.; writing—review and editing, Y.Y., X.X., and X.Z.; supervision, Q.H., Y.K., and Y.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Climate Change Vulnerability Assessment for Urban Trees in Wuhan, Fundamental Research Funds for the Central Universities (2662022YLQD002).

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Brysse, K.; Oreskes, N.; O'Reilly, J.; Oppenheimer, M. Climate Change Prediction: Erring on the Side of Least Drama? *Glob. Environ. Change* **2013**, *23*, 327–337. [\[CrossRef\]](#)
2. Global Change and the Ecology of Cities | Science. Available online: <https://www.science.org/doi/10.1126/science.1150195> (accessed on 8 December 2022).
3. Grimm, N.B.; Faeth, S.H.; Golubiewski, N.E.; Redman, C.L.; Wu, J.; Bai, X.; Briggs, J.M. Global Change and the Ecology of Cities. *Science* **2008**, *319*, 756–760. [\[CrossRef\]](#) [\[PubMed\]](#)
4. Heidari, H.; Mohammadbeigi, A.; Khazaei, S.; Soltsoanzadeh, A.; Asgarian, A.; Saghafipour, A. The Effects of Climatic and Environmental Factors on Heat-Related Illnesses: A Systematic Review from 2000 to 2020. *Urban Clim.* **2020**, *34*, 100720. [\[CrossRef\]](#)
5. Hwang, W.H.; Wiseman, P.E.; Thomas, V.A. Enhancing the Energy Conservation Benefits of Shade Trees in Dense Residential Developments Using an Alternative Tree Placement Strategy. *Landsc. Urban Plan.* **2017**, *158*, 62–74. [\[CrossRef\]](#)
6. Feng, L.; Yang, S.; Zhou, Y.; Shuai, L. Exploring the Effects of the Spatial Arrangement and Leaf Area Density of Trees on Building Wall Temperature. *Build. Environ.* **2021**, *205*, 108295. [\[CrossRef\]](#)
7. Wang, Z.-H.; Zhao, X.; Yang, J.; Song, J. Cooling and Energy Saving Potentials of Shade Trees and Urban Lawns in a Desert City. *Appl. Energy* **2016**, *161*, 437–444. [\[CrossRef\]](#)
8. Zhang, Y.; Murray, A.T.; Turner, B.L. Optimizing Green Space Locations to Reduce Daytime and Nighttime Urban Heat Island Effects in Phoenix, Arizona. *Landsc. Urban Plan.* **2017**, *165*, 162–171. [\[CrossRef\]](#)
9. Livesley, S.J.; McPherson, G.M.; Calfapietra, C. The Urban Forest and Ecosystem Services: Impacts on Urban Water, Heat, and Pollution Cycles at the Tree, Street, and City Scale. *J. Environ. Qual.* **2016**, *45*, 119–124. [\[CrossRef\]](#)
10. Morakinyo, T.E.; Lam, Y.F. Simulation Study on the Impact of Tree-Configuration, Planting Pattern and Wind Condition on Street-Canyon's Micro-Climature and Thermal Comfort. *Build. Environ.* **2016**, *103*, 262–275. [\[CrossRef\]](#)
11. Ryu, Y.-H.; Bou-Zeid, E.; Wang, Z.-H.; Smith, J.A. Realistic Representation of Trees in an Urban Canopy Model. *Bound. Layer Meteorol.* **2016**, *159*, 193–220. [\[CrossRef\]](#)
12. Lai, D.; Liu, W.; Gan, T.; Liu, K.; Chen, Q. A Review of Mitigating Strategies to Improve the Thermal Environment and Thermal Comfort in Urban Outdoor Spaces. *Sci. Total Environ.* **2019**, *661*, 337–353. [\[CrossRef\]](#) [\[PubMed\]](#)
13. Bowler, D.E.; Buyung-Ali, L.; Knight, T.M.; Pullin, A.S. Urban Greening to Cool Towns and Cities: A Systematic Review of the Empirical Evidence. *Landsc. Urban Plan.* **2010**, *97*, 147–155. [\[CrossRef\]](#)
14. Morakinyo, T.E.; Kong, L.; Lau, K.K.-L.; Yuan, C.; Ng, E. A Study on the Impact of Shadow-Cast and Tree Species on in-Canyon and Neighborhood's Thermal Comfort. *Build. Environ.* **2017**, *115*, 1–17. [\[CrossRef\]](#)
15. Kántor, N.; Chen, L.; Gál, C.V. Human-Biometeorological Significance of Shading in Urban Public Spaces—Summertime Measurements in Pécs, Hungary. *Landsc. Urban Plan.* **2018**, *170*, 241–255. [\[CrossRef\]](#)
16. Lee, I.; Voogt, J.A.; Gillespie, T.J. Analysis and Comparison of Shading Strategies to Increase Human Thermal Comfort in Urban Areas. *Atmosphere* **2018**, *9*, 91. [\[CrossRef\]](#)
17. Zhang, J.; Zhang, F.; Gou, Z.; Liu, J. Assessment of Macroclimate and Microclimate Effects on Outdoor Thermal Comfort via Artificial Neural Network Models. *Urban Clim.* **2022**, *42*, 101134. [\[CrossRef\]](#)
18. Zhang, J.; Khoshbakht, M.; Liu, J.; Gou, Z.; Xiong, J.; Jiang, M. A Clustering Review of Vegetation-Indicating Parameters in Urban Thermal Environment Studies towards Various Factors. *J. Therm. Biol.* **2022**, *110*, 103340. [\[CrossRef\]](#) [\[PubMed\]](#)
19. Moher, D.; Liberati, A.; Tetzlaff, J.; Altman, D.G. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *BMJ* **2009**, *339*, b2535. [\[CrossRef\]](#)
20. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews. *BMJ* **2021**, *372*, n71. [\[CrossRef\]](#)
21. Lin, Y.-H.; Tsai, K.-T. Screening of Tree Species for Improving Outdoor Human Thermal Comfort in a Taiwanese City. *Sustainability* **2017**, *9*, 340. [\[CrossRef\]](#)
22. Rahman, M.A.; Hartmann, C.; Moser-Reischl, A.; von Strachwitz, M.F.; Paeth, H.; Pretzsch, H.; Pauleit, S.; Rötzer, T. Tree Cooling Effects and Human Thermal Comfort under Contrasting Species and Sites. *Agric. For. Meteorol.* **2020**, *287*, 107947. [\[CrossRef\]](#)
23. Fang, H.; Baret, F.; Plummer, S.; Schaepman-Strub, G. An Overview of Global Leaf Area Index (LAI): Methods, Products, Validation, and Applications. *Rev. Geophys.* **2019**, *57*, 739–799. [\[CrossRef\]](#)
24. Zhang, J.; Gou, Z.; Zhang, F.; Shutter, L. A Study of Tree Crown Characteristics and Their Cooling Effects in a Subtropical City of Australia. *Ecol. Eng.* **2020**, *158*, 106027. [\[CrossRef\]](#)
25. Morakinyo, T.E.; Lau KK, L.; Ren, C.; Ng, E. Performance of Hong Kong's Common Trees Species for Outdoor Temperature Regulation, Thermal Comfort and Energy Saving. *Build. Environ.* **2018**, *137*, 157–170. [\[CrossRef\]](#)
26. WinSCANOPY: Canopy Structure and Solar Radiation Analysis. Available online: https://regentinstruments.com/assets/winscanopy_about.html (accessed on 23 May 2024).
27. Wei, J.; Li, H.; Wang, Y.; Xu, X. The Cooling and Humidifying Effects and the Thresholds of Plant Community Structure Parameters in Urban Aggregated Green Infrastructure. *Forests* **2021**, *12*, 111. [\[CrossRef\]](#)
28. Kong, F.; Yan, W.; Zheng, G.; Yin, H.; Cavan, G.; Zhan, W.; Zhang, N.; Cheng, L. Retrieval of Three-Dimensional Tree Canopy and Shade Using Terrestrial Laser Scanning (TLS) Data to Analyze the Cooling Effect of Vegetation. *Agric. For. Meteorol.* **2016**, *217*, 22–34. [\[CrossRef\]](#)

29. Feng, L.; Tian, H.; Qiao, Z.; Zhao, M.; Liu, Y. Detailed Variations in Urban Surface Temperatures Exploration Based on Unmanned Aerial Vehicle Thermography. *IEEE J. Sel. Top. Appl. Earth Observ. Remote Sens.* **2020**, *13*, 204–216. [\[CrossRef\]](#)
30. Winbourne, J.B.; Jones, T.S.; Garvey, S.M.; Harrison, J.L.; Wang, L.; Li, D.; Templer, P.H.; Hutya, L.R. Tree Transpiration and Urban Temperatures: Current Understanding, Implications, and Future Research Directions. *BioScience* **2020**, *70*, 576–588. [\[CrossRef\]](#)
31. Kong, L.; Lau, K.K.-L.; Yuan, C.; Chen, Y.; Xu, Y.; Ren, C.; Ng, E. Regulation of Outdoor Thermal Comfort by Trees in Hong Kong. *Sust. Cities Soc.* **2017**, *31*, 12–25. [\[CrossRef\]](#)
32. Makaremi, N.; Salleh, E.; Jaafar, M.Z.; GhaffarianHoseini, A. Thermal Comfort Conditions of Shaded Outdoor Spaces in Hot and Humid Climate of Malaysia. *Build. Environ.* **2012**, *48*, 7–14. [\[CrossRef\]](#)
33. Bröde, P.; Fiala, D.; Błażejczyk, K.; Holmér, I.; Jendritzky, G.; Kampmann, B.; Tinz, B.; Havenith, G. Deriving the Operational Procedure for the Universal Thermal Climate Index (UTCI). *Int. J. Biometeorol.* **2012**, *56*, 481–494. [\[CrossRef\]](#) [\[PubMed\]](#)
34. Middel, A.; Selover, N.; Hagen, B.; Chhetri, N. Impact of Shade on Outdoor Thermal Comfort—A Seasonal Field Study in Tempe, Arizona. *Int. J. Biometeorol.* **2016**, *60*, 1849–1861. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Zhang, J.; Gou, Z. Tree Crowns and Their Associated Summertime Microclimatic Adjustment and Thermal Comfort Improvement in Urban Parks in a Subtropical City of China. *Urban For. Urban Green.* **2021**, *59*, 126912. [\[CrossRef\]](#)
36. Xiong, J.; Cheng, B.; Zhang, J.; Liu, Y.; Tan, X.; Shi, M.; He, X.; Guo, J. A Study of Waterside Microenvironmental Factors and Their Effects on Summer Outdoor Thermal Comfort in a Cfa-Climate Campus. *J. Therm. Biol.* **2023**, *117*, 103700. [\[CrossRef\]](#) [\[PubMed\]](#)
37. Tseliou, A.; Tsiros, I.X.; Nikolopoulou, M. Seasonal Differences in Thermal Sensation in the Outdoor Urban Environment of Mediterranean Climates—The Example of Athens, Greece. *Int. J. Biometeorol.* **2017**, *61*, 1191–1208. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Elnabawi, M.H.; Hamza, N.; Dudek, S. Thermal Perception of Outdoor Urban Spaces in the Hot Arid Region of Cairo, Egypt. *Sustain. Cities Soc.* **2016**, *22*, 136–145. [\[CrossRef\]](#)
39. Yu, Q.; Ji, W.; Pu, R.; Landry, S.; Acheampong, M.; O’Neil-Dunne, J.; Ren, Z.; Tanim, S.H. A Preliminary Exploration of the Cooling Effect of Tree Shade in Urban Landscapes. *Int. J. Appl. Earth Obs. Geoinf.* **2020**, *92*, 102161. [\[CrossRef\]](#)
40. Aleksandrowicz, O.; Zur, S.; Lebendiger, Y.; Lerman, Y. Shade Maps for Prioritizing Municipal Microclimatic Action in Hot Climates: Learning from Tel Aviv-Yafo. *Sustain. Cities Soc.* **2020**, *53*, 101931. [\[CrossRef\]](#)
41. Zeeshan, M.; Ali, Z.; Ranjha, Q.A. Green Infrastructure with Actual Canopy Parameterization: A Simulation Study for Heat Stress Mitigation in a Hot-Humid Urban Environment. *Arboric. Urban For. (AUF)* **2023**, *49*, 247–267. [\[CrossRef\]](#)
42. Shen, X.; Li, J.; Wang, S.; Su, L.; Li, L. The Impact of Tree-Planting Location on the Microclimate and Thermal Comfort of the Micro-Public Space. *Pol. J. Environ. Stud.* **2023**, *32*, 717–730. [\[CrossRef\]](#)
43. Wang, H.; Cai, Y.; Deng, W.; Li, C.; Dong, Y.; Zhou, L.; Sun, J.; Li, C.; Song, B.; Zhang, F.; et al. The Effects of Tree Canopy Structure and Tree Coverage Ratios on Urban Air Temperature Based on ENVI-Met. *Forests* **2023**, *14*, 80. [\[CrossRef\]](#)
44. Chen, H.; Liu, R.; Zhang, Y. The Impact of Vegetation Canopy on the Outdoor Thermal Environment in Cold Winter and Spring. *Sustainability* **2023**, *15*, 12818. [\[CrossRef\]](#)
45. Zhao, Q.; Sailor, D.J.; Wentz, E.A. Impact of Tree Locations and Arrangements on Outdoor Microclimates and Human Thermal Comfort in an Urban Residential Environment. *Urban For. Urban Green.* **2018**, *32*, 81–91. [\[CrossRef\]](#)
46. Jareemit, D.; Srivanit, M. A Comparative Study of Cooling Performance and Thermal Comfort under Street Market Shades and Tree Canopies in Tropical Savanna Climate. *Sustainability* **2022**, *14*, 4653. [\[CrossRef\]](#)
47. Sodoudi, S.; Zhang, H.; Chi, X.; Müller, F.; Li, H. The Influence of Spatial Configuration of Green Areas on Microclimate and Thermal Comfort. *Urban For. Urban Green.* **2018**, *34*, 85–96. [\[CrossRef\]](#)
48. Yilmaz, S.; Kulekci, E.A.; Mutlu, B.E.; Sezen, I. Analysis of Winter Thermal Comfort Conditions: Street Scenarios Using ENVI-Met Model. *Environ. Sci. Pollut. Res.* **2021**, *28*, 63837–63859. [\[CrossRef\]](#) [\[PubMed\]](#)
49. Yang, Q.; Huang, X.; Tong, X.; Xiao, C.; Yang, J.; Liu, Y.; Cao, Y. Global Assessment of Urban Trees’ Cooling Efficiency Based on Satellite Observations. *Environ. Res. Lett.* **2022**, *17*, 034029. [\[CrossRef\]](#)
50. Ziter, C.D.; Pedersen, E.J.; Kucharik, C.J.; Turner, M.G. Scale-Dependent Interactions between Tree Canopy Cover and Impervious Surfaces Reduce Daytime Urban Heat during Summer. *Proc. Natl. Acad. Sci. USA* **2019**, *116*, 7575–7580. [\[CrossRef\]](#) [\[PubMed\]](#)
51. Ryu, Y.-H.; Baik, J.-J. Quantitative Analysis of Factors Contributing to Urban Heat Island Intensity. *J. Appl. Meteorol. Climatol.* **2012**, *51*, 842–854. [\[CrossRef\]](#)
52. Buyantuyev, A.; Wu, J. Urban Heat Islands and Landscape Heterogeneity: Linking Spatiotemporal Variations in Surface Temperatures to Land-Cover and Socioeconomic Patterns. *Landsc. Ecol.* **2010**, *25*, 17–33. [\[CrossRef\]](#)
53. Liu, Y.; Huang, X.; Yang, Q.; Cao, Y. The Turning Point between Urban Vegetation and Artificial Surfaces for Their Competitive Effect on Land Surface Temperature. *J. Clean. Prod.* **2021**, *292*, 126034. [\[CrossRef\]](#)
54. Alonzo, M.; Baker, M.E.; Gao, Y.; Shandas, V. Spatial Configuration and Time of Day Impact the Magnitude of Urban Tree Canopy Cooling. *Environ. Res. Lett.* **2021**, *16*, 084028. [\[CrossRef\]](#)
55. Xu, C.; Chen, G.; Huang, Q.; Su, M.; Rong, Q.; Yue, W.; Haase, D. Can Improving the Spatial Equity of Urban Green Space Mitigate the Effect of Urban Heat Islands? An Empirical Study. *Sci. Total Environ.* **2022**, *841*, 156687. [\[CrossRef\]](#) [\[PubMed\]](#)
56. Wang, W.; Liu, K.; Tang, R.; Wang, S. Remote Sensing Image-Based Analysis of the Urban Heat Island Effect in Shenzhen, China. *Phys. Chem. Earth Parts A/B/C* **2019**, *110*, 168–175. [\[CrossRef\]](#)
57. Imran, H.M.; Kala, J.; Ng, A.W.M.; Muthukumaran, S. Effectiveness of Vegetated Patches as Green Infrastructure in Mitigating Urban Heat Island Effects during a Heatwave Event in the City of Melbourne. *Weather Clim. Extrem.* **2019**, *25*, 100217. [\[CrossRef\]](#)

58. Greene, C.S.; Kedron, P.J. Beyond Fractional Coverage: A Multilevel Approach to Analyzing the Impact of Urban Tree Canopy Structure on Surface Urban Heat Islands. *Appl. Geogr.* **2018**, *95*, 45–53. [\[CrossRef\]](#)
59. Yang, G.; Yu, Z.; Jørgensen, G.; Vejre, H. How Can Urban Blue-Green Space Be Planned for Climate Adaption in High-Latitude Cities? A Seasonal Perspective. *Sustain. Cities Soc.* **2020**, *53*, 101932. [\[CrossRef\]](#)
60. Park, Y.; Guldmann, J.-M.; Liu, D. Impacts of Tree and Building Shades on the Urban Heat Island: Combining Remote Sensing, 3D Digital City and Spatial Regression Approaches. *Comput. Environ. Urban Syst.* **2021**, *88*, 101655. [\[CrossRef\]](#)
61. Fung, C.K.W.; Jim, C.Y. Microclimatic Resilience of Subtropical Woodlands and Urban-Forest Benefits. *Urban For. Urban Green.* **2019**, *42*, 100–112. [\[CrossRef\]](#)
62. Wang, C.; Wang, Z.; Yang, J. Cooling Effect of Urban Trees on the Built Environment of Contiguous United States. *Earth's Future* **2018**, *6*, 1066–1081. [\[CrossRef\]](#)
63. Yan, H.; Wu, F.; Dong, L. Influence of a Large Urban Park on the Local Urban Thermal Environment. *Sci. Total Environ.* **2018**, *622–623*, 882–891. [\[CrossRef\]](#) [\[PubMed\]](#)
64. Andreou, E. Thermal Comfort in Outdoor Spaces and Urban Canyon Microclimate. *Renew. Energy* **2013**, *55*, 182–188. [\[CrossRef\]](#)
65. Manickathan, L.; Defraeye, T.; Allegrini, J.; Derome, D.; Carmeliet, J. Parametric Study of the Influence of Environmental Factors and Tree Properties on the Transpirative Cooling Effect of Trees. *Agric. For. Meteorol.* **2018**, *248*, 259–274. [\[CrossRef\]](#)
66. Li, Z.; Zhang, H.; Juan, Y.-H.; Lee, Y.-T.; Wen, C.-Y.; Yang, A.-S. Effects of Urban Tree Planting on Thermal Comfort and Air Quality in the Street Canyon in a Subtropical Climate. *Sustain. Cities Soc.* **2023**, *91*, 104334. [\[CrossRef\]](#)
67. Morakinyo, T.E.; Ouyang, W.; Lau, K.K.-L.; Ren, C.; Ng, E. Right Tree, Right Place (Urban Canyon): Tree Species Selection Approach for Optimum Urban Heat Mitigation—Development and Evaluation. *Sci. Total Environ.* **2020**, *719*, 137461. [\[CrossRef\]](#) [\[PubMed\]](#)
68. Sabrin, S.; Karimi, M.; Nazari, R.; Pratt, J.; Bryk, J. Effects of Different Urban-Vegetation Morphology on the Canopy-Level Thermal Comfort and the Cooling Benefits of Shade Trees: Case-Study in Philadelphia. *Sustain. Cities Soc.* **2021**, *66*, 102684. [\[CrossRef\]](#)
69. Atwa, S.; Ibrahim, M.G.; Murata, R. Evaluation of Plantation Design Methodology to Improve the Human Thermal Comfort in Hot-Arid Climatic Responsive Open Spaces. *Sustain. Cities Soc.* **2020**, *59*, 102198. [\[CrossRef\]](#)
70. Zhang, L.; Zhan, Q.; Lan, Y. Effects of the Tree Distribution and Species on Outdoor Environment Conditions in a Hot Summer and Cold Winter Zone: A Case Study in Wuhan Residential Quarters. *Build. Environ.* **2018**, *130*, 27–39. [\[CrossRef\]](#)
71. Kántor, N.; Kovács, A.; Takács, Á. Small-Scale Human-Biometeorological Impacts of Shading by a Large Tree. *Open Geosci.* **2016**, *8*, 231–245. [\[CrossRef\]](#)
72. Jiao, M.; Zhou, W.; Zheng, Z.; Yan, J.; Wang, J. Optimizing the Shade Potential of Trees by Accounting for Landscape Context. *Sustain. Cities Soc.* **2021**, *70*, 102905. [\[CrossRef\]](#)
73. Chengping, X.; Chunhui, Y.; Jing, R.; Jiying, L. Numerical Analysis the Effect of Trees on the Outdoor Thermal Environment and the Building Energy Consumption in a Residential Neighborhood. *IOP Conf. Ser. Earth Environ. Sci.* **2020**, *546*, 032007. [\[CrossRef\]](#)
74. Rouhollahi, M.; Whaley, D.; Behrend, M.; Byrne, J.; Boland, J. The Role of Residential Tree Arrangement: A Scoping Review of Energy Efficiency in Temperate to Subtropical Climate Zones. *Renew. Sustain. Energy Rev.* **2022**, *158*, 112155. [\[CrossRef\]](#)
75. Ogueke, N.V.; Nwakanma, A.F.; Ngharamike, T.; Nduka, C.F.; Onyejizu, E.F.; Anyanwu, E.E. Energy-Saving Potentials of Some Local Trees. *Energy Effic.* **2017**, *10*, 171–181. [\[CrossRef\]](#)
76. Palme, M.; Privitera, R.; La Rosa, D. The Shading Effects of Green Infrastructure in Private Residential Areas: Building Performance Simulation to Support Urban Planning. *Energy Build.* **2020**, *229*, 110531. [\[CrossRef\]](#)
77. Simpson, J.R.; McPherson, E.G. Simulation of Tree Shade Impacts on Residential Energy Use for Space Conditioning in Sacramento. *Atmos. Environ.* **1998**, *32*, 69–74. [\[CrossRef\]](#)
78. Huang, Y.; Akbari, H.; Taha, H.; Rosenfeld, A. The Potential of Vegetation in Reducing Summer Cooling Loads in Residential Buildings. *J. Clim. Appl. Meteorol.* **1987**, *26*, 1103–1116. [\[CrossRef\]](#)
79. Zhao, Q.; Wentz, E.A.; Murray, A.T. Tree Shade Coverage Optimization in an Urban Residential Environment. *Build. Environ.* **2017**, *115*, 269–280. [\[CrossRef\]](#)
80. Ko, Y.; Radke, J.D. The Effect of Urban Form and Residential Cooling Energy Use in Sacramento, California. *Environ. Plann. B Plan. Des.* **2014**, *41*, 573–593. [\[CrossRef\]](#)
81. Zhang, X.; Ni, L.; Zhang, T.; Qi, F. Influence of Arbor on the Cooling Load Characteristics of Rural Houses—A Case Study in the Region of Hangzhou. *Sustainability* **2023**, *15*, 6853. [\[CrossRef\]](#)
82. Hsieh, C.-M.; Li, J.-J.; Zhang, L.; Schwegler, B. Effects of Tree Shading and Transpiration on Building Cooling Energy Use. *Energy Build.* **2018**, *159*, 382–397. [\[CrossRef\]](#)
83. Balter, J.; Ganem, C.; Discoli, C. On High-Rise Residential Buildings in an Oasis-City: Thermal and Energy Assessment of Different Envelope Materiality above and below Tree Canopy. *Energy Build.* **2016**, *113*, 61–73. [\[CrossRef\]](#)
84. Li, X.; Ratti, C.; Seiferling, I. Quantifying the Shade Provision of Street Trees in Urban Landscape: A Case Study in Boston, USA, Using Google Street View. *Landsc. Urban Plan.* **2018**, *169*, 81–91. [\[CrossRef\]](#)
85. Zhao, D.; Lei, Q.; Shi, Y.; Wang, M.; Chen, S.; Shah, K.; Ji, W. Role of Species and Planting Configuration on Transpiration and Microclimate for Urban Trees. *Forests* **2020**, *11*, 825. [\[CrossRef\]](#)
86. Streiling, S.; Matzarakis, A. Influence of Single and Small Clusters of Trees on the Bioclimate of A City: A Case Study. *Arboric. Urban For. (AUF)* **2003**, *29*, 309–316. [\[CrossRef\]](#)

87. Richards, D.; Fung, T.; Belcher, R.; Edwards, P. Differential Air Temperature Cooling Performance of Urban Vegetation Types in the Tropics. *Urban For. Urban Green.* **2020**, *50*, 126651. [\[CrossRef\]](#)
88. Lee, H.; Mayer, H.; Chen, L. Contribution of Trees and Grasslands to the Mitigation of Human Heat Stress in a Residential District of Freiburg, Southwest Germany. *Landsc. Urban Plan.* **2016**, *148*, 37–50. [\[CrossRef\]](#)
89. Wang, X. The Influence of Vertical Canopy Structure on the Cooling and Humidifying Urban Microclimate during Hot Summer Days. *Landsc. Urban Plan.* **2023**, *238*, 104841. [\[CrossRef\]](#)
90. Jamei, E.; Rajagopalan, P.; Seyedmahmoudian, M.; Jamei, Y. Review on the Impact of Urban Geometry and Pedestrian Level Greening on Outdoor Thermal Comfort. *Renew. Sustain. Energy Rev.* **2016**, *54*, 1002–1017. [\[CrossRef\]](#)
91. Jiao, M.; Zhou, W.; Zheng, Z.; Wang, J.; Qian, Y. Patch Size of Trees Affects Its Cooling Effectiveness: A Perspective from Shading and Transpiration Processes. *Agric. For. Meteorol.* **2017**, *247*, 293–299. [\[CrossRef\]](#)
92. Wang, X.; Dallimer, M.; Scott, C.E.; Shi, W.; Gao, J. Tree Species Richness and Diversity Predicts the Magnitude of Urban Heat Island Mitigation Effects of Greenspaces. *Sci. Total Environ.* **2021**, *770*, 145211. [\[CrossRef\]](#)
93. Wujeska-Klaue, A.; Pfautsch, S. The Best Urban Trees for Daytime Cooling Leave Nights Slightly Warmer. *Forests* **2020**, *11*, 945. [\[CrossRef\]](#)
94. Zhang, Z.; Lv, Y.; Pan, H. Cooling and Humidifying Effect of Plant Communities in Subtropical Urban Parks. *Urban For. Urban Green.* **2013**, *12*, 323–329. [\[CrossRef\]](#)
95. Cai, Y.; Li, C.; Ye, L.; Xiao, L.; Gao, X.; Mo, L.; Du, H.; Zhou, Y.; Zhou, G. Effect of the Roadside Tree Canopy Structure and the Surrounding on the Daytime Urban Air Temperature in Summer. *Agric. For. Meteorol.* **2022**, *316*, 108850. [\[CrossRef\]](#)
96. Chen, T.; Yang, H.; Chen, G.; Lam, C.K.C.; Hang, J.; Wang, X.; Liu, Y.; Ling, H. Integrated Impacts of Tree Planting and Aspect Ratios on Thermal Environment in Street Canyons by Scaled Outdoor Experiments. *Sci. Total Environ.* **2021**, *764*, 142920. [\[CrossRef\]](#) [\[PubMed\]](#)
97. Chinchilla, J.; Carbonnel, A.; Galleguillos, M. Effect of Urban Tree Diversity and Condition on Surface Temperature at the City Block Scale. *Urban For. Urban Green.* **2021**, *60*, 127069. [\[CrossRef\]](#)
98. Rendon, P.; Love, N.; Pawlak, C.; Yost, J.; Ritter, M.; Doremus, J. Street Tree Diversity and Urban Heat. *Urban For. Urban Green.* **2024**, *91*, 128180. [\[CrossRef\]](#)
99. Helletsgruber, C.; Gillner, S.; Gulyás, Á.; Junker, R.R.; Tanács, E.; Hof, A. Identifying Tree Traits for Cooling Urban Heat Islands—A Cross-City Empirical Analysis. *Forests* **2020**, *11*, 1064. [\[CrossRef\]](#)
100. Wang, J.; Guo, W.; Wang, C.; Yao, Y.; Kou, K.; Xian, D.; Zhang, Y. Tree Crown Geometry and Its Performances on Human Thermal Comfort Adjustment. *J. Urban Manag.* **2021**, *10*, 16–26. [\[CrossRef\]](#)
101. Liu, H.; Lim, J.Y.; Wint Hnin Thet, B.; Lai, P.-Y.; Koh, W.S. Evaluating the Impact of Tree Morphologies and Planting Densities on Outdoor Thermal Comfort in Tropical Residential Precincts in Singapore. *Build. Environ.* **2022**, *221*, 109268. [\[CrossRef\]](#)
102. Zhang, X.; Lei, Y.; Li, R.; Ackerman, A.; Guo, N.; Li, Y.; Yang, Q.; Liu, Y. Research on Thermal Comfort of Underside of Street Tree Based on LiDAR Point Cloud Model. *Forests* **2022**, *13*, 1086. [\[CrossRef\]](#)
103. Smithers, R.J.; Doick, K.J.; Burton, A.; Sibille, R.; Steinbach, D.; Harris, R.; Groves, L.; Blicharska, M. Comparing the Relative Abilities of Tree Species to Cool the Urban Environment. *Urban Ecosyst.* **2018**, *21*, 851–862. [\[CrossRef\]](#)
104. Rahman, M.A.; Stratopoulos, L.M.F.; Moser-Reischl, A.; Zölch, T.; Häberle, K.-H.; Rötzer, T.; Pretzsch, H.; Pauleit, S. Traits of Trees for Cooling Urban Heat Islands: A Meta-Analysis. *Build. Environ.* **2020**, *170*, 106606. [\[CrossRef\]](#)
105. England, J.R.; Attiwill, P.M. Changes in Leaf Morphology and Anatomy with Tree Age and Height in the Broadleaved Evergreen Species, *Eucalyptus Regnans* F. Muell. *Trees* **2006**, *20*, 79–90. [\[CrossRef\]](#)
106. Zheng, S.; Guldman, J.-M.; Liu, Z.; Zhao, L. Influence of Trees on the Outdoor Thermal Environment in Subtropical Areas: An Experimental Study in Guangzhou, China. *Sustain. Cities Soc.* **2018**, *42*, 482–497. [\[CrossRef\]](#)
107. Wang, Z.; Li, Y.; Song, J.; Wang, K.; Xie, J.; Chan, P.W.; Ren, C.; Di Sabatino, S. Modelling and Optimizing Tree Planning for Urban Climate in a Subtropical High-Density City. *Urban Clim.* **2022**, *43*, 101141. [\[CrossRef\]](#)
108. Stratopoulos, L.M.F.; Duthweiler, S.; Häberle, K.-H.; Pauleit, S. Effect of Native Habitat on the Cooling Ability of Six Nursery-Grown Tree Species and Cultivars for Future Roadside Plantings. *Urban For. Urban Green.* **2018**, *30*, 37–45. [\[CrossRef\]](#)
109. Gillner, S.; Vogt, J.; Tharang, A.; Dettmann, S.; Roloff, A. Role of Street Trees in Mitigating Effects of Heat and Drought at Highly Sealed Urban Sites. *Landsc. Urban Plan.* **2015**, *143*, 33–42. [\[CrossRef\]](#)
110. Chen, X.; Zhao, P.; Hu, Y.; Ouyang, L.; Zhu, L.; Ni, G. Canopy Transpiration and Its Cooling Effect of Three Urban Tree Species in a Subtropical City—Guangzhou, China. *Urban For. Urban Green.* **2019**, *43*, 126368. [\[CrossRef\]](#)
111. Rahman, M.A.; Dervishi, V.; Moser-Reischl, A.; Ludwig, F.; Pretzsch, H.; Rötzer, T.; Pauleit, S. Comparative Analysis of Shade and Underlying Surfaces on Cooling Effect. *Urban For. Urban Green.* **2021**, *63*, 127223. [\[CrossRef\]](#)
112. Herb, W.R.; Janke, B.; Mohseni, O.; Stefan, H.G. Ground Surface Temperature Simulation for Different Land Covers. *J. Hydrol.* **2008**, *356*, 327–343. [\[CrossRef\]](#)
113. Rahman, M.A.; Moser, A.; Rötzer, T.; Pauleit, S. Comparing the Transpirational and Shading Effects of Two Contrasting Urban Tree Species. *Urban Ecosyst.* **2019**, *22*, 683–697. [\[CrossRef\]](#)
114. Shahidan, M.F.; Jones, P.J.; Gwilliam, J.; Salleh, E. An Evaluation of Outdoor and Building Environment Cooling Achieved through Combination Modification of Trees with Ground Materials. *Build. Environ.* **2012**, *58*, 245–257. [\[CrossRef\]](#)
115. Chen, Y.; Zheng, B.; Hu, Y. Numerical Simulation of Local Climate Zone Cooling Achieved through Modification of Trees, Albedo and Green Roofs—A Case Study of Changsha, China. *Sustainability* **2020**, *12*, 2752. [\[CrossRef\]](#)

116. Wang, C.; Wang, Z.-H.; Kaloush, K.E.; Shacat, J. Cool Pavements for Urban Heat Island Mitigation: A Synthetic Review. *Renew. Sustain. Energy Rev.* **2021**, *146*, 111171. [[CrossRef](#)]
117. Yang, F.; Lau, S.S.Y.; Qian, F. Thermal Comfort Effects of Urban Design Strategies in High-Rise Urban Environments in a Sub-Tropical Climate. *Arch. Sci. Rev.* **2011**, *54*, 285–304. [[CrossRef](#)]
118. Gaffin, S.R.; Rosenzweig, C.; Kong, A.Y.Y. Adapting to Climate Change through Urban Green Infrastructure. *Nat. Clim. Chang.* **2012**, *2*, 704. [[CrossRef](#)]
119. Zhou, W.; Wang, J.; Cadenasso, M.L. Effects of the Spatial Configuration of Trees on Urban Heat Mitigation: A Comparative Study. *Remote Sens. Environ.* **2017**, *195*, 1–12. [[CrossRef](#)]
120. Yu, Z.; Xu, S.; Zhang, Y.; Jørgensen, G.; Vejre, H. Strong Contributions of Local Background Climate to the Cooling Effect of Urban Green Vegetation. *Sci. Rep.* **2018**, *8*, 6798. [[CrossRef](#)] [[PubMed](#)]
121. Krayenhoff, E.S.; Jiang, T.; Christen, A.; Martilli, A.; Oke, T.R.; Bailey, B.N.; Nazarian, N.; Voogt, J.A.; Giometto, M.G.; Stastny, A.; et al. A Multi-Layer Urban Canopy Meteorological Model with Trees (BEP-Tree): Street Tree Impacts on Pedestrian-Level Climate. *Urban Clim.* **2020**, *32*, 100590. [[CrossRef](#)]
122. Segura, R.; Krayenhoff, E.S.; Martilli, A.; Badia, A.; Estruch, C.; Ventura, S.; Villalba, G. How Do Street Trees Affect Urban Temperatures and Radiation Exchange? Observations and Numerical Evaluation in a Highly Compact City. *Urban Clim.* **2022**, *46*, 101288. [[CrossRef](#)]
123. Sanusi, R.; Johnstone, D.; May, P.; Livesley, S.J. Microclimate Benefits That Different Street Tree Species Provide to Sidewalk Pedestrians Relate to Differences in Plant Area Index. *Landsc. Urban Plan.* **2017**, *157*, 502–511. [[CrossRef](#)]
124. De Quadros, B.M.; Mizgier, M.G.O. Urban Green Infrastructures to Improve Pedestrian Thermal Comfort: A Systematic Review. *Urban For. Urban Green.* **2023**, *88*, 128091. [[CrossRef](#)]
125. Wright, A.J.; Francia, R.M. Plant Traits, Microclimate Temperature and Humidity: A Research Agenda for Advancing Nature-based Solutions to a Warming and Drying Climate. *J. Ecol.* **2024**, *1365–2745*, 14313. [[CrossRef](#)]
126. Teskey, R.; Wertin, T.; Bauweraerts, I.; Ameye, M.; McGuire, M.A.; Steppe, K. Responses of Tree Species to Heat Waves and Extreme Heat Events. *Plant Cell Environ.* **2015**, *38*, 1699–1712. [[CrossRef](#)] [[PubMed](#)]
127. Allen, C.D.; Breshears, D.D.; McDowell, N.G. On Underestimation of Global Vulnerability to Tree Mortality and Forest Die-off from Hotter Drought in the Anthropocene. *Ecosphere* **2015**, *6*, 129. [[CrossRef](#)]
128. Hauer, R.J.; Hanou, I.S.; Sivyer, D. Planning for Active Management of Future Invasive Pests Affecting Urban Forests: The Ecological and Economic Effects of Varying Dutch Elm Disease Management Practices for Street Trees in Milwaukee, WI USA. *Urban Ecosyst.* **2020**, *23*, 1005–1022. [[CrossRef](#)]
129. Das, S.; Sun, Q.; Zhou, H. GeoAI to Implement an Individual Tree Inventory: Framework and Application of Heat Mitigation. *Urban For. Urban Green.* **2022**, *74*, 127634. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.